

IKS4001 – Mathematics in India

Assignment 2 – Complete Solutions

Indian Mathematical Heritage: Aryabhata, Vedic Mathematics & Number Theory

S'O'A Deemed to be University, Bhubaneswar

Q1. Who was Aryabhata? Briefly explain his contributions to mathematics.

About Aryabhata

Aryabhata (476–550 CE) was one of the greatest mathematicians and astronomers of ancient India. He was born in Kusumapura (modern-day Patna, Bihar) and is believed to have also worked in Taregana (near Patna). He is the first Indian mathematician-astronomer known by name, and his work marks the beginning of the classical age of Indian mathematics.

His seminal work, the Aryabhatiya (499 CE), composed when he was only 23 years old, is a concise treatise covering arithmetic, algebra, plane and spherical trigonometry, and mathematical astronomy.

Major Mathematical Contributions

- Value of π (Pi): He gave an approximate value of π as $62832/20000 = 3.1416$ — accurate to four decimal places, and noted it was 'approximate' (āsanna), indicating awareness of irrationality.
- Trigonometry: Introduced the concept of the half-chord (ardha-jyā), which later became the modern 'sine'. Constructed a table of 24 sine values at intervals of $3^\circ 45'$.
- Kuttaka Method: Developed the 'Pulverizer' algorithm for solving linear Diophantine equations — a precursor to the extended Euclidean algorithm.
- Square and Cube Roots: Gave systematic methods for extracting square and cube roots.
- Arithmetic and Geometric Progressions: Formulas for sums of natural numbers, squares, and cubes: $1^2+2^2+\dots+n^2 = n(n+1)(2n+1)/6$.
- Algebra: Solved quadratic and indeterminate equations.
- Positional Value System: Used a positional notation system with a zero concept.
- Astronomy: Correctly explained rotation of the earth on its axis; gave periods of planets; calculated the length of a sidereal year as 365 days 6 hours 12 minutes 30 seconds (extremely accurate).

Q2. Describe the structure of the Aryabhatiya and explain its four chapters.

The Aryabhatiya is a compact astronomical treatise containing 121 verses in Sanskrit. It is divided into four sections called padas (quarters or chapters):

Chapter	Name	Verses	Content
1	Gitikapada	13 verses	Large units of time, cosmology, planetary revolutions, table of sines (jyā values for 0° – 90°), and introduction of large number terminology.
2	Ganitapada	33 verses	Core mathematics: arithmetic, geometric and algebraic rules, area & volume of geometric figures, arithmetic progressions, square/cube roots, shadow problems, and the value of π .
3	Kalakriyapada	25 verses	Units of time (day, month, year, yuga), celestial mechanics, planetary motions, intercalary

			months, and the calculation of mean and true positions of planets.
4	Golapada	50 verses	Spherical astronomy: the celestial sphere, ecliptic, celestial equator, shape of the earth, causes of eclipses, horizon, rising and setting of planets and stars.

The Gitikapada uses a special metrical encoding where numerical values are encoded in syllables — an innovative data-compression technique. The Ganitapada is mathematically most significant. The Golapada (Sphere chapter) is the most extensive, reflecting Aryabhata's emphasis on three-dimensional astronomy.

Q3. What is meant by 'Siddhanta'? Who wrote the Pañcasiddhāntikā? Name the five Siddhantas.

Meaning of Siddhanta

The word Siddhanta (Sanskrit: सिद्धान्त) means 'established doctrine' or 'proven principle'. In the context of Indian astronomy and mathematics, a Siddhanta is a comprehensive treatise that presents a complete, self-consistent astronomical system, including mathematical calculations for planetary motions, time-keeping, calendar systems, spherical astronomy, and cosmology. It represents the culmination of systematic scientific inquiry.

Author of Pañcasiddhāntikā

The Pañcasiddhāntikā ('Five Astronomical Canons') was written by Varāhamihira (c. 505 CE), a celebrated Indian scholar, mathematician and astronomer at the court of the Gupta king Vikramāditya II. The text is a summary and critical comparison of five existing astronomical traditions (siddhantas) of his time.

The Five Siddhantas

No.	Siddhanta	Key Features
1	Paulisha Siddhanta	Based on Greek (Paulisa = Paul of Alexandria) astronomical tradition; uses Greek-style epicyclic models; among the most accurate of the five.
2	Romaka Siddhanta	Influenced by Roman (Romaka = Roman) astronomical knowledge; uses a luni-solar cycle of 2850 years; connects Indian and Hellenistic traditions.
3	Vasishtha Siddhanta	Attributed to the sage Vasishtha; based on the Vedic tradition; older text with simplified planetary models.
4	Saura (Surya) Siddhanta	Attributed to the Sun god (Surya); one of the most authoritative texts in Indian astronomy; still used in Panchangam calculations today; contains precise astronomical data.
5	Paitamaha Siddhanta	Attributed to Brahma (Pitamaha = Grandfather); among the oldest; uses Vedic cosmological framework with a Kalpa-based yuga system.

Q4. What is the Kuttaka (Pulverizer) method developed by Aryabhata? Explain its purpose.

Definition and Origin

The Kuttaka (Sanskrit: कुट्टक, meaning 'to grind' or 'to pulverize') is an algorithmic method developed by Aryabhata for solving linear indeterminate equations — equations with multiple unknowns that require integer solutions. It is equivalent to, and historically precedes, what we today call the Extended Euclidean Algorithm.

Purpose and Problem Type

The Kuttaka solves equations of the form:

$$ax \equiv c \pmod{b} \quad \text{or equivalently} \quad ax - by = c$$

where a, b, c are known integers and x, y are integer unknowns. Such equations arise in astronomy (computing the position of planets, synchronizing calendars) and number theory.

The Kuttaka Algorithm

The method works by 'grinding' the coefficients through successive division until a trivially solvable equation is reached, then back-substituting. The steps are:

- Step 1: Use Euclidean-style repeated division: Divide a by b to get remainder r_1 : $a = q_1b + r_1$.
- Step 2: Continue dividing: $b = q_2r_1 + r_2$, $r_1 = q_3r_2 + r_3$, ... until remainder = 1 (if $\gcd(a,b) = 1$).
- Step 3: Check feasibility: The equation $ax - by = c$ has integer solutions if and only if $\gcd(a,b) \mid c$.
- Step 4: Build the solution back by reverse substitution — the 'pulverizing' that gives the method its name.
- Step 5: The general solution: if (x_0, y_0) is one solution, all solutions are $x = x_0 + bt$, $y = y_0 + at$ for integer t .

The Kuttaka was instrumental in Aryabhata's astronomical computations — for example, in determining when multiple planetary cycles coincide (conjunction problems). It predates the work of Brahmagupta and is considered the world's first systematic algorithm for solving Diophantine equations.

Q5. Explain Aryabhata's rule for the value of π accurate to four decimal places.

In the Ganitapada (verse 10) of the Aryabhataiya, Aryabhata states:

*"caturadhikam śatamaṣṭaguṇam dvāṣaṣṭistathā sahasrāṇām |
ayutadvaya viṣkambhasy āsanno vṛttapariṇāhaḥ ||"*

Translation: 'Add 4 to 100, multiply by 8, and add 62000. The result is approximately (āsanna) the circumference of a circle whose diameter is 20000.'

Mathematical Derivation

Following Aryabhata's rule step by step:

$$\text{Step 1: } 100 + 4 = 104$$

$$\text{Step 2: } 104 \times 8 = 832$$

$$\text{Step 3: } 832 + 62000 = 62832$$

$$\text{Circumference} = 62832 \text{ for Diameter} = 20000$$

$$\text{Therefore: } \pi = C/D = 62832 / 20000 = 3.1416$$

The actual value of $\pi = 3.14159265\dots$, so Aryabhata's value 3.1416 is correct to four decimal places (error $< 0.01\%$).

Importantly, Aryabhata used the word *āsanna* ('approximate'), showing he was aware that this was not the exact value — a remarkable early recognition of the irrationality of π .

Aryabhata's Value	Actual Value of π
$62832 / 20000 = 3.1416$	3.14159265358979...
Error ≈ 0.0002 ($< 0.01\%$)	Ratio: $3.1416 / 3.14159 \approx 1.00000031$

Q6. Explain the Polygon Doubling Formula for Computing π .

The polygon doubling method (attributed also to Archimedes, 287–212 BCE, but described in various Indian texts) computes π by successively doubling the number of sides of a regular polygon inscribed in (or circumscribed about) a unit circle, with the polygon's perimeter approaching 2π .

Core Idea

If a regular polygon with n sides is inscribed in a circle of radius R , its perimeter $= n \times s_n$, where s_n is the side length. As $n \rightarrow \infty$, the perimeter $\rightarrow 2\pi R$.

The Doubling Formula

If s_n is the side length of a regular n -gon inscribed in a unit circle ($R = 1$), then the side length of a $2n$ -gon is:

$$s_{2n} = \sqrt{2 - \sqrt{4 - s_n^2}}$$

This formula comes from the half-angle relationship in trigonometry: $\sin(\theta/2) = \sqrt{(1 - \cos\theta)/2}$, and using the Pythagorean theorem on the chord.

Step-by-step Computation (starting from regular hexagon)

Start: Regular hexagon inscribed in unit circle ($R = 1$). Each side $= R = 1$ (since the side of a hexagon equals the radius).

n (sides)	s_n (side length)	Perimeter $= n \cdot s_n$	$\pi \approx \text{Perimeter}/2$
6	1.000000	6.000000	3.000000
12	0.517638	6.211657	3.105828
24	0.261052	6.265257	3.132629

48	0.130806	6.278700	3.139350
96	0.065438	6.282064	3.141032
192	0.032723	6.282905	3.141453
∞ (limit)	$\rightarrow 0$	$\rightarrow 2\pi$	$\rightarrow \pi = 3.14159\dots$

Aryabhata likely used a version of this method to arrive at his value 3.1416. The polygon doubling approach connects directly to his sine difference formula, which effectively computes chord lengths iteratively.

Q7. Describe Aryabhata's Recursive Difference Sine Difference Formula.

Aryabhata presented a remarkable recursive formula for computing sine differences in Ganitapada verses 11–12 of the Aryabhatiya. He divided the quadrant (0° to 90°) into 24 equal parts of $3^\circ 45'$ (= 225 arc-minutes) each.

The Formula

Let $\Delta_1 = D_1 = \sin(\alpha)$, where $\alpha = 3^\circ 45'$. Define the n -th sine difference as:

$$D_n = \text{jya}(n\alpha) - \text{jya}((n-1)\alpha) \quad [\text{where } \text{jya} = R \cdot \sin, R = 3438]$$

Aryabhata's recursive difference formula (in modern notation) is:

$$D_{n+1} = D_n - D_n \cdot D_1 / \text{jya}(\alpha)$$

Or equivalently, using second differences:

$$D_{n+1} = D_n - \text{jya}(n\alpha) / 225$$

where $225 = R \cdot \sin(\alpha) / \sin(\alpha) \approx 3438 / 15.24$ (Aryabhata's specific constant with $R = 3438$ and $\alpha = 3^\circ 45'$).

Key Properties

- It reduces the computation of all 24 sine values to just repeated addition and subtraction.
- The second differences of sines are proportional to the sine values themselves — a discrete analogue of $d^2(\sin x)/dx^2 = -\sin x$.
- Starting values: $D_1 = \text{jya}(3^\circ 45') = 225$ (in Aryabhata's units where $R = 3438$).

First few values ($R = 3438$)

n	Angle $n\alpha$	D_n (difference)	$\text{jya}(n\alpha)$	Modern $\sin(n\alpha) \times 3438$
1	$3^\circ 45'$	225	225	224.86
2	$7^\circ 30'$	224	449	448.75
3	$11^\circ 15'$	222	671	670.72
6	$22^\circ 30'$	210	1315	1305.68
24	$90^\circ 00'$	—	3438	3438.00 (= R)

Q8. Discuss Aryabhata's Contribution to Trigonometry and Sine Tables.

Introduction of the Sine Concept

Aryabhata is credited with introducing the trigonometric concept of the 'half-chord' (ardha-jyā → shortened to jyā → transliterated to Arabic as 'jiba' → later misread as 'jaib' → translated to Latin as 'sinus'). This is the direct etymological ancestor of the modern word 'sine'. Before Aryabhata, Indian astronomers used the full-chord (jyā); his innovation of using half-chords aligned the concept with what we now call the sine function.

The Aryabhata Sine Table

In the Gitikapada, Aryabhata encoded 24 sine values (in his own numerical code, the āryabhaṭīya notation) for angles from 3°45' to 90°, using a radius R = 3438 (approximately equal to the number of arc minutes in a radian: $180 \times 60 / \pi \approx 3437.75$).

n	Angle	Aryabhata's jyā	R·sin(angle)	Error
1	3° 45'	225	224.86	0.14
2	7° 30'	449	448.75	0.25
3	11° 15'	671	670.72	0.28
4	15° 00'	890	889.82	0.18
6	22° 30'	1315	1305.68	9.32
12	45° 00'	2431	2430.94	0.06
24	90° 00'	3438	3438.00	0.00

Significance

- Aryabhata's table was the first systematic, high-precision trigonometric table in Indian mathematics.
- His recursive difference formula (Q7) made the computation efficient — only additions and one division were needed.
- The table was later used by Brahmagupta, Bhaskara I (who gave a rational approximation for sine), and transmitted to Islamic astronomers via Al-Biruni's translation of the Aryabhaṭīya.
- It influenced the development of trigonometry worldwide through its Arabic translation and eventual dissemination to Europe.

Q9. Calculate cube roots of non-perfect numbers using Vedic mathematics.

Vedic Method (Newton's First-Approximation)

For a non-perfect cube N, find the nearest integer a such that a^3 is closest to N:

$$\sqrt[3]{N} \approx a + (N - a^3) / (3a^2) \quad [\text{if } N > a^3, \text{ approach from below}]$$
$$\sqrt[3]{N} \approx a - (a^3 - N) / (3a^2) \quad [\text{if } N < a^3, \text{ approach from above}]$$

Use the NEARER of the two surrounding perfect cubes for better accuracy:

N	Near cube	a	Calculation	Approx. $\sqrt[3]{N}$	Actual $\sqrt[3]{N}$
29	$27 = 3^3$	3	$3 + (29-27)/(3 \times 9) = 3 + 2/27$	≈ 3.074	3.072
38	$27 = 3^3$	3	$3 + (38-27)/(3 \times 9) = 3 + 11/27$	≈ 3.407	3.362
53	$64 = 4^3$	4	$4 - (64-53)/(3 \times 16) = 4 - 11/48$	≈ 3.771	3.756
59	$64 = 4^3$	4	$4 - (64-59)/(3 \times 16) = 4 - 5/48$	≈ 3.896	3.893
67	$64 = 4^3$	4	$4 + (67-64)/(3 \times 16) = 4 + 3/48$	≈ 4.063	4.062
79	$64 = 4^3$	4	$4 + (79-64)/(3 \times 16) = 4 + 15/48$	≈ 4.313	4.291
107	$125 = 5^3$	5	$5 - (125-107)/(3 \times 25) = 5 - 18/75$	≈ 4.760	4.747
131	$125 = 5^3$	5	$5 + (131-125)/(3 \times 25) = 5 + 6/75$	≈ 5.080	5.079

Q10. Calculate cube roots of perfect numbers using Vedic mathematics.

Vedic Cube Root Method for Perfect Cubes

Step 1: Group digits in sets of three from the right (e.g., 9261 $\rightarrow$ 9 | 261).

Step 2: The leftmost group determines the first digit (largest $n^3 \leq$ leftmost group).

Step 3: The units digit of the cube root is determined by the units digit of the number using the Vedic 'last digit' table:

Units digit of N^3	Units digit of $\sqrt[3]{N}$
0	0
1	1
2	8
3	7
4	4
5	5
6	6
7	3
8	2
9	9

Solutions

Number	Groups	1st digit	Last digit reasoning	Answer
9261	9 261	2 ($2^3=8\leq 9$)	Ends in 1 $\rightarrow$ root ends in 1; try 21: $21^3=9261$ $\checkmark$	21
74088	74 088	4 ($4^3=64\leq 74$)	Ends in 8 $\rightarrow$ root ends in 2; try 42: $42^3=74088$ $\checkmark$	42
6859	6 859	1 ($1^3=1\leq 6$)	Ends in 9 $\rightarrow$ root ends in 9; try 19: $19^3=6859$ $\checkmark$	19
50653	50 653	3 ($3^3=27\leq 50$)	Ends in 3 $\rightarrow$ root ends in 7; try 37: $37^3=50653$ $\checkmark$	37
389017	389 017	7 ($7^3=343\leq 389$)	Ends in 7 $\rightarrow$ root ends in 3; try 73: $73^3=389017$ $\checkmark$	73
42875	42 875	3 ($3^3=27\leq 42$)	Ends in 5 $\rightarrow$ root ends in 5; try 35: $35^3=42875$ $\checkmark$	35
-941192	941 192	9 ($9^3=729\leq 941$)	Ends in 2 $\rightarrow$ root ends in 8; $\sqrt[3]{941192} = 98$; $98^3=941192$ $\checkmark$	-98
-226981	226 981	6 ($6^3=216\leq 226$)	Ends in 1 $\rightarrow$ root ends in 1; $\sqrt[3]{226981} = 61$; $61^3=226981$ $\checkmark$	-61
438.976	438976/1000	$\sqrt[3]{438976} = 76$	438976: 438 976 $\rightarrow$ $7^3=343\leq 438$; ends in 6 $\rightarrow$ root ends 6; $76^3=438976$ $\checkmark$ $\rightarrow$ $\div \sqrt[3]{1000}=10$	7.6
-531.441	531441/1000	$\sqrt[3]{531441} = 81$	531 441 $\rightarrow$ $8^3=512\leq 531$; ends in 1 $\rightarrow$ root ends 1; $81^3=531441$ $\checkmark$ $\rightarrow$ $\div 10$, negate	-8.1

Q11. Find square roots using Vedic mathematics (long division method).

Vedic Square Root: Digit-by-Digit Long Division

Pair digits from the decimal point (left for integer part, right for decimal part). For each pair, find the largest digit d such that $(2 \times \text{result_so_far} \times 10 + d) \times d \leq \text{current_remainder}$.

i. $\sqrt{2376.592}$

Group: (23)(76).(59)(20)

Step 1: Group 23 $\rightarrow 4^2=16\leq 23$. First digit = 4. Remainder = $23-16 = 7$.

Step 2: Bring down 76 $\rightarrow 776$. Trial: $2 \times 4 = 8$. Find d : $(80+d) \times d \leq 776$. $d=8$: $88 \times 8 = 704$ $\checkmark$. Remainder = $776-704 = 72$. Result: 48.

Step 3: Bring down 59 $\rightarrow 7259$. Trial: $2 \times 48 = 96$. Find d : $(960+d) \times d \leq 7259$. $d=7$: $967 \times 7 = 6769$ $\checkmark$. Remainder = $7259-6769 = 490$. Result: 48.7.

Step 4: Bring down 20 → 49020. Trial: $2 \times 487 = 974$. $d=5$: $9745 \times 5 = 48725$ ✓. Result: 48.75.

$$\sqrt{2376.592} \approx 48.75 \quad [\text{Check: } 48.75^2 = 2376.5625 \approx 2376.592 \quad \checkmark]$$

ii. $\sqrt{15876}$

Group: (1)(58)(76). Check: $126^2 = 15876$.

Step 1: Group 1 → $1^2=1$. First digit = 1. Remainder = 0.

Step 2: Bring down 58 → 058. Trial: $2 \times 1 = 2$. $d=2$: $22 \times 2 = 44$ ✓. Remainder = $58 - 44 = 14$. Result: 12.

Step 3: Bring down 76 → 1476. Trial: $2 \times 12 = 24$. $d=6$: $246 \times 6 = 1476$ ✓. Remainder = 0.

$$\sqrt{15876} = 126 \quad [\text{Check: } 126^2 = 15876 \quad \checkmark \text{ (Perfect Square)}]$$

iii. $\sqrt{125316}$

Group: (12)(53)(16). Check: $354^2 = 125316$.

Step 1: Group 12 → $3^2=9 \leq 12$. First digit = 3. Remainder = $12 - 9 = 3$.

Step 2: Bring down 53 → 353. Trial: $2 \times 3 = 6$. $d=5$: $65 \times 5 = 325$ ✓. Remainder = $353 - 325 = 28$. Result: 35.

Step 3: Bring down 16 → 2816. Trial: $2 \times 35 = 70$. $d=4$: $704 \times 4 = 2816$ ✓. Remainder = 0.

$$\sqrt{125316} = 354 \quad [\text{Check: } 354^2 = 125316 \quad \checkmark \text{ (Perfect Square)}]$$

iv. $\sqrt{3.58021}$

Group from decimal: (03).(58)(02)(10)

Step 1: Group 03 → $1^2=1 \leq 3$. First digit = 1. Remainder = $3 - 1 = 2$.

Step 2: Bring down 58 → 258. Trial: $2 \times 1 = 2$. $d=8$: $28 \times 8 = 224$ ✓. Remainder = $258 - 224 = 34$. Result: 1.8.

Step 3: Bring down 02 → 3402. Trial: $2 \times 18 = 36$. $d=9$: $369 \times 9 = 3321$ ✓. Remainder = $3402 - 3321 = 81$. Result: 1.89.

Step 4: Bring down 10 → 8110. Trial: $2 \times 189 = 378$. $d=2$: $3782 \times 2 = 7564$ ✓. Result: 1.892.

$$\sqrt{3.58021} \approx 1.892 \quad [\text{Check: } 1.892^2 = 3.5797 \approx 3.580 \quad \checkmark]$$

v. $\sqrt{0.8931}$

Group from decimal: .(89)(31)(00)

Step 1: Group 89 → $9^2=81 \leq 89$. First digit = 0.9. Remainder = $89 - 81 = 8$.

Step 2: Bring down 31 → 831. Trial: $2 \times 9 = 18$. $d=4$: $184 \times 4 = 736$ ✓. Remainder = $831 - 736 = 95$. Result: 0.94.

Step 3: Bring down 00 → 9500. Trial: $2 \times 94 = 188$. $d=5$: $1885 \times 5 = 9425$ ✓. Remainder = 75. Result: 0.945.

$$\sqrt{0.8931} \approx 0.945 \quad [\text{Check: } 0.945^2 = 0.893025 \approx 0.8931 \quad \checkmark]$$

vi. $\sqrt{372.4318}$

Group: (03)(72).(43)(18)

Step 1: Group 03 → $1^2=1 \leq 3$. First digit = 1. Remainder = 2.

Step 2: Bring down 72 → 272. $d=9$: $29 \times 9 = 261$ ✓. Remainder = 11. Result: 19.

Step 3: Bring down 43 → 1143. Trial: $2 \times 19 = 38$. $d=2$: $382 \times 2 = 764$ ✓. Remainder = 379. Result: 19.2.

Step 4: Bring down 18 → 37918. Trial: $2 \times 192 = 384$. $d=9$: $3849 \times 9 = 34641$ ✓. Remainder = 3277. Result: 19.29.

Step 5: Continue → Result: 19.30.

$$\sqrt{372.4318} \approx 19.30 \quad [\text{Check: } 19.30^2 = 372.49 \approx 372.43 \text{ ✓}]$$

Q12. Kuttaka Method (Euclidean Algorithm) for Linear Diophantine Equation.

Note: The exact equation in the original assignment was not fully visible in the PDF. Below, we demonstrate the Kuttaka method using a standard example: solve $17x - 5y = 1$ (find integer solutions), which is a classic illustration used in Indian mathematical texts.

Example: Solve $17x - 5y = 1$

Step 1: Check solvability. $\gcd(17, 5) = 1$, which divides 1. So integer solutions exist.

Step 2: Apply the Euclidean Algorithm (the 'pulverizing' step):

$$\begin{aligned} 17 &= 3 \times 5 + 2 && \rightarrow \text{quotient } q_1 = 3, \text{ remainder } r_1 = 2 \\ 5 &= 2 \times 2 + 1 && \rightarrow \text{quotient } q_2 = 2, \text{ remainder } r_2 = 1 \\ 2 &= 2 \times 1 + 0 && \rightarrow \gcd = 1 \end{aligned}$$

Step 3: Back-substitute (Kuttaka reverse):

$$\begin{aligned} 1 &= 5 - 2 \times 2 \\ 2 &= 17 - 3 \times 5 \\ 1 &= 5 - 2 \times (17 - 3 \times 5) \\ 1 &= 5 - 2 \times 17 + 6 \times 5 \\ 1 &= 7 \times 5 - 2 \times 17 \\ \therefore 17 \times (-2) - 5 \times (-7) &= 1 \quad \rightarrow \text{one solution: } x_0 = -2, y_0 = -7 \end{aligned}$$

Step 4: General Solution:

$$\begin{aligned} x &= -2 + 5t \\ y &= -7 + 17t \quad \text{for any integer } t \end{aligned}$$

Verification ($t = 1$): $x = 3, y = 10$. $17 \times 3 - 5 \times 10 = 51 - 50 = 1$ ✓

Q13. Linear Diophantine Equations: General Solutions.

Note: The original equations in the PDF were partially cut off. Below, representative equations consistent with the course level are solved. The number '47' visible in the PDF suggests one equation involves 47.

(a) $47x + 30y = 1$

gcd(47, 30): Apply Euclidean algorithm:

$$47 = 1 \times 30 + 17$$

$$30 = 1 \times 17 + 13$$

$$17 = 1 \times 13 + 4$$

$$13 = 3 \times 4 + 1$$

$$4 = 4 \times 1 + 0 \rightarrow \gcd(47, 30) = 1$$

Since gcd = 1 divides 1, solutions exist. Back-substituting:

$$1 = 13 - 3 \times 4 = 13 - 3 \times (17 - 13) = 4 \times 13 - 3 \times 17$$

$$= 4 \times (30 - 17) - 3 \times 17 = 4 \times 30 - 7 \times 17$$

$$= 4 \times 30 - 7 \times (47 - 30) = 11 \times 30 - 7 \times 47$$

$$\therefore 47 \times (-7) + 30 \times (11) = 1 \rightarrow x_0 = -7, y_0 = 11$$

$$\text{General: } x = -7 + 30t, \quad y = 11 - 47t \quad (t \in \mathbb{Z})$$

(b) $14x + 21y = 35$

Simplify: gcd(14, 21) = 7. Since $7 \mid 35$, solutions exist. Divide through by 7: $2x + 3y = 5$.

$$\gcd(2, 3) = 1. \text{ Euclidean: } 3 = 1 \times 2 + 1 \rightarrow 1 = 3 - 1 \times 2$$

$$\therefore 2 \times (-1) + 3 \times (1) = 1 \rightarrow \text{multiply by 5: } 2 \times (-5) + 3 \times 5 = 5$$

$$x_0 = -5, y_0 = 5 \quad (\text{for simplified equation } 2x + 3y = 5)$$

$$\text{General: } x = -5 + 3t, \quad y = 5 - 2t \quad (t \in \mathbb{Z})$$

Verification ($t=2$): $x = 1, y = 1$. $14 \times 1 + 21 \times 1 = 35 \checkmark$

(c) $6x + 4y = 3$

gcd(6, 4) = 2. Since 2 does NOT divide 3, NO INTEGER SOLUTIONS EXIST.

Proof: $6x + 4y = 2(3x + 2y)$ is always even, but 3 is odd.
Contradiction.

Q14. Ekanyunena Purvena: 'By one less than the previous one'.

The Ekanyunena Purvena sutra is used for multiplying a number by a repunit of 9s (99, 999, 9999, etc.). The rule: for $N \times (10^n - 1)$:

$$\text{Result} = (N - 1) \text{ concatenated with } (10^n - N)$$

Equivalently: Left part = $N-1$; Right part = Complement of N (nines complement for n digits).

i. 9999×2378

Here $9999 = 10^4 - 1$, and the multiplier has 4 digits (same as number of 9s).

$$\text{Left part: } 2378 - 1 = 2377$$

$$\text{Right part: } 9999 - 2377 = 7622$$

$$\text{Result: } 2377 \mid 7622 = 23777622$$

Verification: $9999 \times 2378 = 10000 \times 2378 - 2378 = 23780000 - 2378 = 23777622 \checkmark$

ii. 999×324567

Here $999 = 10^3 - 1$, but 324567 has 6 digits > 3 (more digits than 9s). Apply as:

$$999 \times 324567 = 324567 \times (1000 - 1) = 324567000 - 324567 = 324242433$$

Vedic shortcut: Split 324567 into two 3-digit halves: 324 and 567.

$$\text{Left part: } 324567 - 1 = 324566$$

$$\text{Right part (nines complement of 324567 using } 10^6): 10^6 - 324567 - \dots$$

$$\text{Direct: } 324567000 - 324567 = 324242433$$

Verification: $1000 \times 324567 = 324567000$; minus 324567 = 324242433 ✓

Q15. Anurupyena (Proportionately).

Note: The specific numerical examples in the original PDF were not visible (cut off). The Anurupyena sutra is presented below with standard illustrative examples used in Vedic Mathematics courses.

The sutra means 'Proportionately'. It is used to simplify multiplication or division by first adjusting to a convenient base (like 10, 100, 50, 25), performing the computation, then correcting proportionally.

Example 1: Multiply 48×46 (base 50 = $100/2$)

Step 1: Compute using base 50. Deviations: $48 - 50 = -2$; $46 - 50 = -4$.

$$\text{Cross-add: } 48 - 4 = 44 \quad (\text{or } 46 - 2 = 44)$$

$$\text{Product of deviations: } (-2)(-4) = 8$$

$$\text{At base 100: result would be } 44|08 = 4408$$

$$\text{Adjust for base 50 (= } 100/2): \text{ Divide left part by 2: } 44 \div 2 = 22$$

$$\text{Answer: } 22|08 = 2208$$

Verification: $48 \times 46 = 2208$ ✓

Example 2: Multiply 32×34 (base 30 = 3×10)

Deviations from 30: +2 and +4.

$$\text{Cross-sum: } 32 + 4 = 36$$

$$\text{Product of deviations: } 2 \times 4 = 8$$

$$\text{At base 10: } 36 \times 3 = 108; \text{ append } 8 = 1088$$

Verification: $32 \times 34 = 1088$ ✓

Example 3: Multiply 204×206 (base 200 = 2×100)

Deviations from 200: +4 and +6.

$$\text{Cross-sum: } 204 + 6 = 210$$

$$\text{Product: } 4 \times 6 = 24$$

$$\text{At base 100: } 210 \times 2 = 420; \text{ append } 24 = 42024$$

Verification: $204 \times 206 = 42024$ ✓

Q16. Compute squares using Vedic Dwandwa (Duplex) method.

Dwandwa (Duplex) Sutra

The Duplex $D(n)$ of a number is computed as:

For a single digit d : $D(d) = d^2$. For two digits pq : $D(pq) = 2pq$. For three digits pqr : $D(pqr) = 2pr + q^2$.

The square is assembled right-to-left, one digit per duplex computation.

i. 149^2

Digits: 1, 4, 9. Compute duplexes right to left:

$$\begin{aligned}
 D(9) &= 81 && \rightarrow \text{write 1, carry 8} \\
 D(49) &= 2 \times 4 \times 9 = 72; +8 = 80 && \rightarrow \text{write 0, carry 8} \\
 D(149) &= 2 \times 1 \times 9 + 4^2 = 18 + 16 = 34; +8 = 42 && \rightarrow \text{write 2, carry 4} \\
 D(14) &= 2 \times 1 \times 4 = 8; +4 = 12 && \rightarrow \text{write 2, carry 1} \\
 D(1) &= 1; +1 = 2 && \rightarrow \text{write 2} \\
 149^2 &= 22201 \quad [\text{Check: } 150^2 - 2 \times 150 + 1 = 22500 - 300 + 1 = 22201 \checkmark]
 \end{aligned}$$

ii. 642^2

$$\begin{aligned}
 D(2) &= 4 && \rightarrow \text{write 4, carry 0} \\
 D(42) &= 2 \times 4 \times 2 = 16 && \rightarrow \text{write 6, carry 1} \\
 D(642) &= 2 \times 6 \times 2 + 4^2 = 24 + 16 = 40; +1 = 41 && \rightarrow \text{write 1, carry 4} \\
 D(64) &= 2 \times 6 \times 4 = 48; +4 = 52 && \rightarrow \text{write 2, carry 5} \\
 D(6) &= 36; +5 = 41 && \rightarrow \text{write 41} \\
 642^2 &= 412164 \quad [\text{Check: } 640^2 + 2 \times 640 \times 2 + 4 = 409600 + 2560 + 4 = 412164 \checkmark]
 \end{aligned}$$

iii. 1024^2

Using Yavadunam (sutra): 1024 is near base 1000.

$$\begin{aligned}
 1024 &= 1000 + 24 \\
 (1000 + 24)^2 &= 1000^2 + 2 \times 1000 \times 24 + 24^2 \\
 &= 1000000 + 48000 + 576 \\
 1024^2 &= 1048576 \quad [\text{Known: } 2^{10} = 1024, 2^{20} = 1048576 \checkmark]
 \end{aligned}$$

iv. 1777^2

1777 is near 1800.

$$\begin{aligned}
 1777 &= 1800 - 23 \\
 (1800 - 23)^2 &= 1800^2 - 2 \times 1800 \times 23 + 23^2 \\
 &= 3240000 - 82800 + 529 \\
 1777^2 &= 3157729 \quad [\text{Check: } 1777 \times 1777 = 3157729 \checkmark]
 \end{aligned}$$

v. 9614^2

9614 is near 10000.

$$\begin{aligned}9614 &= 10000 - 386 \\(10000 - 386)^2 &= 10000^2 - 2 \times 10000 \times 386 + 386^2 \\&= 100000000 - 7720000 + 148996 \\9614^2 &= 92428996 \quad [\text{Check: } (9600+14)^2 = 92160000+268800+196 = \\&92428996 \checkmark]\end{aligned}$$

Q17. Find positive square roots using Vedic mathematics.

Method: Identify perfect squares by digit-pattern recognition + Vedic verification

i. $\sqrt{390625}$

Note the last two digits: 25 $\rightarrow$ square root ends in 5 (since $n^2 \rightarrow 25$ means n ends in 5).

Groups: (3)(90)(6)(25). Estimate: $600^2 = 360000$, $700^2 = 490000$. Try 625.

$$\begin{aligned}625^2 &= (600+25)^2 = 360000 + 30000 + 625 = 390625 \checkmark \\ \sqrt{390625} &= 625\end{aligned}$$

ii. $\sqrt{1234321}$

Observe the symmetric digit pattern: 1-2-3-4-3-2-1. This is characteristic of $(1111)^2$.

$$\begin{aligned}1111^2 &= (1100+11)^2 = 1210000 + 24200 + 121 = 1234321 \checkmark \\ \sqrt{1234321} &= 1111\end{aligned}$$

iii. $\sqrt{285156}$

Ends in 6 $\rightarrow$ root ends in 4 or 6. Estimate: $500^2 = 250000$, $550^2 = 302500$. Try 534.

$$\begin{aligned}534^2 &= (530+4)^2 = 280900 + 4240 + 16 = 285156 \checkmark \\ \sqrt{285156} &= 534\end{aligned}$$

iv. $\sqrt{31329}$

Ends in 9 $\rightarrow$ root ends in 3 or 7. Estimate: $170^2 = 28900$, $180^2 = 32400$. Try 177.

$$\begin{aligned}177^2 &= (170+7)^2 = 28900 + 2380 + 49 = 31329 \checkmark \\ \sqrt{31329} &= 177\end{aligned}$$

v. $\sqrt{135424}$

Ends in 4 $\rightarrow$ root ends in 2 or 8. Estimate: $360^2 = 129600$, $380^2 = 144400$. Try 368.

$$368^2 = (370-2)^2 = 136900 - 1480 + 4 = 135424 \checkmark$$

$$\sqrt{135424} = 368$$

Number	Square Root	Verification
390625	625	$625^2 = 390625 \checkmark$
1234321	1111	$1111^2 = 1234321 \checkmark$
285156	534	$534^2 = 285156 \checkmark$
31329	177	$177^2 = 31329 \checkmark$
135424	368	$368^2 = 135424 \checkmark$

Q18. Names of large numbers mentioned in Vedic literature.

Ancient Indian texts — including the Yajurveda, Ramayana, Mahabharata, and Jain canonical texts — contain elaborate naming systems for very large numbers, extending far beyond what any contemporary civilization had conceived. This reflects the Indian philosophical tradition of dealing with cosmic time scales and astronomical distances.

Vedic & Classical Number Names

Power	Number	Sanskrit Name	Source / Notes
10^7	10,000,000	Koti / Crore	Yajurveda, Aryabhatiya; still in daily use in India as 'crore'.
10^8	100,000,000	Arbuda	Yajurveda (Taittiriya Samhita); Atharvaveda.
10^9	1,000,000,000	Abja / Padma	Abja = 'water-born'; Padma = lotus. Used in Puranas for cosmic quantities.
10^{10}	10,000,000,000	Kharva	Yajurveda (Vajasaneyi Samhita 17.2); cosmic scale calculations.
10^{11}	100,000,000,000	Nikharva	Vedic literature; 'ni-kharva' = below kharva (counting downward in cosmic scales).
10^{12}	1,000,000,000,000	Mahapadma / Shanku	Mahapadma = great lotus; used in Ramayana for Ravana's army count.
10^{14}	100,000,000,000,000	Samudra / Antya	Samudra = ocean; Antya = the final/ultimate; Valmiki Ramayana (Yuddha Kanda, Sarga 28).
10^{16}	10,000,000,000,000,000	Madhya / Parardha	Madhya = middle; some texts give Parardha = 10^{16} or 10^{17} depending on the system.

10^{17}	100,000,000,000,000,000	Parardha	Parardha = 'beyond half'; highest named number in many Vedic systems; Taittiriya Samhita.
10^{53}	10^{53}	Tallakshana	From Jain canonical texts (Anuyogadvara Sutra); Jain mathematicians extended to astronomically large numbers for describing cycles of rebirth.

Commentary

The existence of systematic names for numbers up to 10^{53} (and beyond in Jain texts) demonstrates that ancient Indian mathematicians had developed a positional, base-10 understanding of arbitrarily large numbers. This stands in sharp contrast to, for example, the ancient Greek numerical system, which effectively stopped at the 'myriad' (10^4). The Vedic naming system reflects both astronomical calculation needs (yuga cycles, cosmic distances) and the philosophical notion that the universe is infinite and quantifiable.

The system also appears in Valmiki's Ramayana (Yuddha Kanda) where Vibhishana enumerates Ravana's army using numbers like Koti, Shankha, Maha-shankha, Vrnda, Maha-vrnda, Padma, Maha-padma, Kharva, Maha-kharva, Samudra, and Ogha — each ten times the previous.

— End of Assignment 2 Solutions —